

HYDROCARBONS

CLASS 11 CHEMISTRY — NEET MASTER NOTES

Color	Meaning	When to Focus
RED (Bold)	Very Important — Must Memorize	Always — NEET guaranteed
BLUE	Medium Important — Should Know	Revise before exam
BLACK	Normal — Background Info	Read once

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CHAPTER OVERVIEW

NEET Scoring Intelligence Map

NEET SCORING MAP

Topic	NEET Frequency	Marks	Priority
Markovnikov / Anti-Markovnikov	Very High	4-6	RED
EAS + Directive Influence (o/m/p)	Very High	4-6	RED
Ozonolysis Products	High	2-4	RED
Free Radical Halogenation Mechanism	High	2-4	RED
Aromaticity + Huckel Rule	High	2-4	BLUE
Geometrical Isomerism (cis/trans)	High	2	BLUE
IUPAC Nomenclature	High	2	BLUE
Lindlar vs Na/liq.NH ₃	High	2	RED
Conformations of Ethane	Medium	2	BLUE
Acidic Character of Alkynes	Medium	2	BLUE
Named Reactions (Wurtz/Kolbe/Decarboxylation)	Medium	2	BLUE

TOP TRAP ZONES — MOST COMMON WRONG ANSWERS IN NEET

Halogens (Cl, Br, F, I) are ORTHO/PARA directors — NOT meta. Students write meta = WRONG.

Anti-Markovnikov (peroxide effect) works ONLY for HBr. Not HCl, not HI.

Lindlar catalyst = CIS alkene. Na/liquid NH₃ = TRANS alkene. Students swap these.

Methane CANNOT be prepared by Kolbe electrolysis.

Staggered conformation = MORE stable (not eclipsed).

trans isomer has HIGHER melting point; cis isomer has HIGHER boiling point.

C-C bond in benzene = 139 pm (not 133 pm, not 154 pm).

Internal alkynes (but-2-yne) do NOT react with NaNH₂. Only terminal alkynes.

Hydration of ethyne gives ETHANAL (not ethanol) — via vinyl alcohol tautomerism.

BHC (Gammmaxane) = ADDITION reaction of benzene + Cl₂ under UV. NOT substitution.

ALKANES

Saturated Hydrocarbons — C_nH_{2n+2} | sp^3 | Paraffins

QUICK FACTS

Property	Value
General Formula	C_nH_{2n+2}
Other Name	Paraffins (parum affinis = little affinity in Latin)
Hybridisation	sp^3 on all carbons
Bond Angle	109.5 degrees (H-C-H)
C-C Bond Length	154 pm
C-H Bond Length	112 pm
First Member	CH ₄ (Methane) — found in coal mines and marshy places
Physical State (298K)	C ₁ -C ₄ : gases C ₅ -C ₁₇ : liquids C ₁₈ +: solids
Solubility	Non-polar; insoluble in water; soluble in non-polar solvents
Reactivity	Inert to acids, bases, oxidising/reducing agents under normal conditions

ISOMERISM IN ALKANES

Chain Isomers: Same molecular formula, different carbon chain structure

Formula	No. of Isomers	Names
C ₄ H ₁₀	2	n-Butane (b.p.273K) 2-Methylpropane/isobutane (b.p.261K)
C ₅ H ₁₂	3	n-Pentane 2-Methylbutane (isopentane) 2,2-Dimethylpropane (neopentane)
C ₆ H ₁₄	5	n-Hexane + 4 branched isomers
C ₇ H ₁₆	9	—
C ₁₀ H ₂₂	75	—

Higher branching = LOWER boiling point

Reason: More branching → more spherical shape → less surface area contact → weaker van der Waals forces

Example: neopentane (b.p. 282.5K) < n-pentane (b.p. 309.1K)

Carbon Types

1° (Primary): Attached to 1 carbon | 2° (Secondary): Attached to 2 carbons

3° (Tertiary): Attached to 3 carbons | 4° (Quaternary): Attached to 4 carbons

PREPARATION OF ALKANES

1. Hydrogenation of Alkenes/Alkynes



Pt/Pd: catalyse at room temperature | Ni: requires higher temperature and pressure

2. Reduction of Alkyl Halides (Zn + dil. HCl)



Works for Cl, Br, I — NOT for fluorides

3. Wurtz Reaction [NEET: High Frequency]



ONLY useful for EVEN carbon alkanes using the SAME alkyl halide

TRAP: Using two DIFFERENT alkyl halides gives mixture of R-R, R-R' and R'-R' — not selective

4. Decarboxylation [NEET: High Frequency]



Product has ONE CARBON LESS than the parent carboxylic acid

Soda lime = NaOH + CaO mixture

TRAP: Sodium propanoate (3C) gives ETHANE (2C) — not propane

5. Kolbe's Electrolytic Method [NEET: High Frequency]



Gives EVEN carbon alkanes at the anode

TRAP: Methane CANNOT be made by Kolbe method — formate gives only CO₂, no CH₃ radical

CHEMICAL PROPERTIES OF ALKANES

1. Free Radical Halogenation [NEET: Very High Frequency]

Reactivity of halogens: $F_2 > Cl_2 > Br_2 > I_2$

Reactivity of H atoms replaced: $3^\circ > 2^\circ > 1^\circ$

F₂: too violent (uncontrollable) | I₂: too slow, reversible — needs HIO₃ or HNO₃

Mechanism — Chlorination of Methane

Step	Name	Reaction	Key Point
1	INITIATION	$Cl-Cl \xrightarrow{[h\nu]} 2 Cl(.)$ (homolysis)	Cl-Cl bond weakest; breaks first by light/heat
2a	PROPAGATION	$CH_4 + Cl(.) \rightarrow (.)CH_3 + HCl$	Methyl free radical formed
2b	PROPAGATION	$(.)CH_3 + Cl_2 \rightarrow CH_3Cl + Cl(.)$	Chain reaction continues
3a	TERMINATION	$Cl(.) + Cl(.) \rightarrow Cl_2$	Radicals recombine
3b	TERMINATION	$(.)CH_3 + (.)CH_3 \rightarrow C_2H_6$	ETHANE byproduct formed here
3c	TERMINATION	$(.)CH_3 + Cl(.) \rightarrow CH_3Cl$	Product formed but chain ends

IMPORTANT: Ethane is a BYPRODUCT of methane chlorination due to termination step 3b

2. Combustion



Incomplete combustion --> Carbon black (used in ink, printer ink, pigments, filters)

3. Controlled Oxidation

Reaction	Condition	Product
$2\text{CH}_4 + \text{O}_2$	Cu, 523K, 100 atm	$2\text{CH}_3\text{OH}$ (Methanol)
$\text{CH}_4 + \text{O}_2$	Mo_2O_3 , heat	$\text{HCHO} + \text{H}_2\text{O}$ (Methanal/Formaldehyde)
$2\text{CH}_3\text{CH}_3 + 3\text{O}_2$	$(\text{CH}_3\text{COO})_2\text{Mn}$, heat	$2\text{CH}_3\text{COOH} + 2\text{H}_2\text{O}$ (Ethanoic acid)
$(\text{CH}_3)_3\text{CH} + \text{KMnO}_4$	Oxidation	$(\text{CH}_3)_3\text{COH}$ (tertiary H only)

4. Isomerisation

n-Hexane $\xrightarrow{[\text{Anhy. AlCl}_3 / \text{HCl}]}$ 2-Methylpentane + 3-Methylpentane

5. Aromatization (Reforming)

n-Hexane $\xrightarrow{[\text{Cr}_2\text{O}_3 \text{ or } \text{V}_2\text{O}_5, \text{Al}_2\text{O}_3, 773\text{K}, 10-20 \text{ atm}]}$ Benzene + 3H_2

n-Hexane (6C) $\rightarrow$ Benzene | n-Heptane (7C) $\rightarrow$ Toluene

6. Pyrolysis / Cracking

Higher alkane $\xrightarrow{[\text{high temp}]}$ lower alkanes + alkenes $\text{C}_{12}\text{H}_{26} \xrightarrow{[\text{Pt/Ni}, 973\text{K}]}$ $\text{C}_7\text{H}_{16} + \text{C}_5\text{H}_{10}$

7. Reaction with Steam

$\text{CH}_4 + \text{H}_2\text{O} \xrightarrow{[\text{Ni}, 1273\text{K}]}$ $\text{CO} + 3\text{H}_2$ (Industrial H_2 production)

CONFORMATIONS OF ETHANE [NEET: Medium Frequency]

Conformation	Dihedral Angle	Torsional Strain	Stability	Energy
Staggered	60°	MINIMUM	MOST STABLE	Lowest
Eclipsed	0°	MAXIMUM	LEAST STABLE	Highest (+12.5 kJ/mol)
Skew/Gauche	$0^\circ-60^\circ$	Intermediate	Intermediate	Middle

Staggered = H atoms farthest apart = minimum repulsion = MOST STABLE

Energy barrier between staggered and eclipsed = 12.5 kJ/mol (very small, so rotation is practically free)

TRAP: Bond angles and bond lengths are SAME in all conformations. Only spatial arrangement changes.

Sawhorse projection: C-C drawn as long line. Front carbon at bottom, rear carbon at top.

Newman projection: View head-on along C-C axis. Front C = dot; rear C = circle.

ALKENES

Unsaturated Hydrocarbons — C_nH_{2n} | sp^2 | Olefins

QUICK FACTS

Property	Value
General Formula	C_nH_{2n} (same as cycloalkanes)
Other Name	Olefins (oil forming — $C_2H_4 + Cl_2$ gives oily liquid)
Hybridisation	sp^2 (doubly bonded carbons)
Bond Angle	About 120° around double bond carbons
C=C Bond Length	134 pm (shorter than C-C: 154 pm)
C=C Bond Enthalpy	681 kJ/mol (sigma ~ 397 + pi ~ 284 kJ/mol)
First Stable Member	Ethene (C_2H_4) also called Ethylene
Physical State	C_2-C_4 : gases C_5-C_{18} : liquids C_{19+} : solids
Nature	Source of loosely held pi electrons — attacked by electrophiles

STRUCTURE OF C=C DOUBLE BOND

C=C = one sigma bond (strong, 397 kJ/mol) + one pi bond (weak, 284 kJ/mol)

Sigma bond: head-on overlap of sp^2 orbitals

Pi bond: lateral/sideways overlap of 2p orbitals (above and below plane)

Pi bond restricts rotation --> cause of geometrical (cis-trans) isomerism

All 6 atoms in ethene ($2C + 4H$) lie in the SAME PLANE (planar molecule)

ISOMERISM IN ALKENES

A. Structural Isomerism

C_4H_8 Isomers	Structure	Type
But-1-ene	$CH_2=CH-CH_2-CH_3$	Position isomer of but-2-ene
But-2-ene	$CH_3-CH=CH-CH_3$	Position isomer of but-1-ene
2-Methylprop-1-ene	$CH_2=C(CH_3)-CH_3$	Chain isomer of above two

B. Geometrical (cis-trans) Isomerism [NEET: High Frequency]

Condition: BOTH doubly bonded carbons must have TWO DIFFERENT substituents

Feature	cis Isomer	trans Isomer
Group arrangement	Same groups on SAME side of C=C	Same groups on OPPOSITE sides
Dipole moment	Higher (e.g. cis-but-2-ene = 0.33D)	Lower/Zero (trans-but-2-ene ~ 0)
Boiling point	HIGHER (more polar)	LOWER

Feature	cis Isomer	trans Isomer
Melting point	LOWER	HIGHER (better crystal packing)
Stability	Less stable (steric strain)	More stable
Example	cis-But-2-ene (b.p. 277K)	trans-But-2-ene (b.p. 274K)

NO cis-trans isomerism if: one doubly bonded C has IDENTICAL groups (e.g. CH₂= or =CR₂)

Example: (CH₃)₂C=CH-C₂H₅ — left carbon has 2 CH₃ (identical) --> NO geometrical isomerism

PREPARATION OF ALKENES

1. Partial Hydrogenation of Alkynes [NEET: Very High]

$\text{RC}\equiv\text{CR} + \text{H}_2 \xrightarrow{\text{Lindlar catalyst}} \text{cis-Alkene}$ $\text{RC}\equiv\text{CR} + \text{H}_2 \xrightarrow{[\text{Na} / \text{liq. NH}_3]} \text{trans-Alkene}$

Lindlar catalyst = Pd/C + quinoline or sulphur compound (partially deactivated)

Lindlar --> CIS alkene | Na/liq.NH₃ --> TRANS alkene

Most repeated NEET trap: Students swap Lindlar and Na/NH₃ products

2. Dehydrohalogenation of Alkyl Halides (alc. KOH)

$\text{R-CH}_2\text{-CH(X)} \xrightarrow{[\text{alc. KOH, heat}]} \text{R-CH=CH}_2 + \text{HX}$ (Beta-elimination: H removed from beta carbon)

Rate: I > Br > Cl (halogen) | tertiary > secondary > primary (alkyl group)

3. Dehalogenation of Vicinal Dihalides (Zn)

$\text{CH}_2\text{Br-CH}_2\text{Br} + \text{Zn} \rightarrow \text{CH}_2=\text{CH}_2 + \text{ZnBr}_2$ (Dehalogenation)

Vicinal = halogen on adjacent (1,2) carbon atoms

4. Acidic Dehydration of Alcohols

$\text{R-CH}_2\text{-CH}_2\text{OH} \xrightarrow{[\text{conc. H}_2\text{SO}_4, 170^\circ\text{C}]} \text{R-CH=CH}_2 + \text{H}_2\text{O}$

Beta-elimination: OH takes the beta-H and leaves as water

CHEMICAL PROPERTIES OF ALKENES

1. Addition of H₂ (Hydrogenation)

$\text{CH}_2=\text{CH}_2 + \text{H}_2 \xrightarrow{[\text{Pt/Pd/Ni}]} \text{CH}_3\text{-CH}_3$

2. Addition of Halogens

$\text{CH}_2=\text{CH}_2 + \text{Br}_2 \xrightarrow{[\text{CCl}_4]} \text{CH}_2\text{Br-CH}_2\text{Br}$ (1,2-Dibromoethane)

REDDISH ORANGE colour of Br₂/CCl₄ DECOLOURISES = TEST FOR UNSATURATION

Iodine does NOT add to alkenes under normal conditions

3. MARKOVNIKOV'S RULE [NEET: Very High — Most Important]

Rule: Negative part (X-) adds to carbon with FEWER H atoms (more substituted)

H+ adds to carbon with MORE H atoms

Reactants	Condition	Major Product	Why
CH ₃ -CH=CH ₂ + HBr	No peroxide	CH ₃ -CHBr-CH ₃ (2-Bromopropane)	2° carbocation more stable than 1°
CH ₃ -CH=CH ₂ + HCl	No peroxide	CH ₃ -CHCl-CH ₃ (2-Chloropropane)	Same logic

Reactants	Condition	Major Product	Why
$\text{CH}_2=\text{CH}_2 + \text{HBr}$	No peroxide	$\text{CH}_3\text{-CH}_2\text{Br}$ (only product)	Symmetric alkene, only one product

Mechanism: H^+ attacks $\text{C}=\text{C} \rightarrow$ more stable 2° carbocation forms (not 1°) $\rightarrow$ Br^- attacks $\rightarrow$ 2-bromopropane

4. ANTI-MARKOVNIKOV (Peroxide / Kharash Effect) [NEET: Very High]

Works **ONLY** for **HBr** in presence of peroxide. **NOT HCl, NOT HI.**

$\text{CH}_3\text{-CH}=\text{CH}_2 + \text{HBr} \xrightarrow{\text{ROOR peroxide}} \text{CH}_3\text{-CH}_2\text{-CH}_2\text{Br}$ (1-Bromopropane)

Br adds to LESS substituted carbon (anti-Markovnikov) via FREE RADICAL mechanism

Step	Reaction	Note
Initiation	$\text{ROOR} \rightarrow 2 \text{RO}(\cdot) \mid \text{RO}(\cdot) + \text{HBr} \rightarrow \text{ROH} + \text{Br}(\cdot)$	Peroxide generates Br radical
Propagation 1	$\text{Br}(\cdot) + \text{CH}_3\text{CH}=\text{CH}_2 \rightarrow \text{CH}_3\text{-CH}(\cdot)\text{-CH}_2\text{Br}$	Br adds; 2° radical more stable than 1°
Propagation 2	$\text{CH}_3\text{-CH}(\cdot)\text{-CH}_2\text{Br} + \text{HBr} \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{Br} + \text{Br}(\cdot)$ H adds to 2° radical end	
Result	1-Bromopropane = major product	Anti-Markovnikov product

HCl: H-Cl bond (430 kJ/mol) too strong for radical cleavage -- no peroxide effect

HI: I radicals recombine to form I_2 instead of adding to double bond

5. Addition of H_2SO_4

$\text{CH}_2=\text{CH}_2 + \text{H}_2\text{SO}_4$ (cold conc.) $\rightarrow \text{CH}_3\text{-CH}_2\text{-OSO}_3\text{H}$ (Ethyl hydrogen sulphate)

Follows Markovnikov rule. Electrophilic addition.

6. Addition of Water (Hydration)

$\text{CH}_3\text{-C}(\text{CH}_3)=\text{CH}_2 + \text{H}_2\text{O} \xrightarrow{[\text{H}^+]} \text{CH}_3\text{-C}(\text{CH}_3)(\text{OH})\text{-CH}_3$ (2-Methylpropan-2-ol) Follows Markovnikov rule

7. Oxidation [NEET: High Frequency]

Reagent	Condition	Product	Test
Cold dil. KMnO_4 (alkaline)	273K, Baeyer's reagent	Vicinal diol (glycol) $\text{CH}_2=\text{CH}_2 \rightarrow \text{HOCH}_2\text{-CH}_2\text{OH}$	Pink KMnO_4 decolourises = unsaturation test
$\text{KMnO}_4 / \text{H}^+$ (acidic, hot)	Hot acidic	Ketones and/or carboxylic acids	—
$\text{CH}_3\text{-CH}=\text{CH-CH}_3 + \text{KMnO}_4$ (acidic)	Hot acidic	2 CH_3COOH (Ethanoic acid)	—

8. OZONOLYSIS [NEET: Very High — Most Important]

Rule: Ozonolysis cleaves $\text{C}=\text{C}$ bond. Each carbon of $\text{C}=\text{C}$ becomes a CARBONYL carbon.

Reagent: O_3 then $\text{Zn}/\text{H}_2\text{O}$ (reductive workup) $\rightarrow$ aldehydes/ketones

Alkene	Products (after $\text{Zn}/\text{H}_2\text{O}$)	Types
$\text{CH}_3\text{-CH}=\text{CH}_2$ (Propene)	CH_3CHO (Ethanal) + HCHO (Methanal)	Aldehyde + Aldehyde
$(\text{CH}_3)_2\text{C}=\text{CH}_2$ (2-Methylpropene)	$(\text{CH}_3)_2\text{C}=\text{O}$ (Propanone) + HCHO	Ketone + Aldehyde
$(\text{CH}_3)_2\text{C}=\text{C}(\text{CH}_3)_2$	2 x $(\text{CH}_3)_2\text{C}=\text{O}$ (Propanone)	Ketone + Ketone
$\text{CH}_3\text{-CH}=\text{CH-CH}_3$ (But-2-ene)	2 x CH_3CHO (Ethanal)	Aldehyde + Aldehyde

SHORTCUT: Terminal $=\text{CH}_2 \rightarrow \text{HCHO}$ (formaldehyde). Internal $=\text{CHR} \rightarrow \text{RCHO}$ (aldehyde). $=\text{CR}_2 \rightarrow$ ketone

9. Polymerisation

$n(\text{CH}_2=\text{CH}_2) \xrightarrow{\text{[high T, high P, catalyst]}} -(\text{CH}_2-\text{CH}_2)_n-$ (Polythene)

$n(\text{CH}_3-\text{CH}=\text{CH}_2) \xrightarrow{\text{[catalyst]}} -(\text{CH}(\text{CH}_3)-\text{CH}_2)_n-$ (Polypropene)

Polythene: bags, bottles, pipes | Polypropene: buckets, milk crates

ALKYNES

Unsaturated Hydrocarbons — C_nH_{2n-2} | sp | Linear

QUICK FACTS

Property	Value
General Formula	C_nH_{2n-2}
Hybridisation	sp (triple bonded carbons)
Bond Angle	180° (H-C≡C-H is linear)
C≡C Bond Length	120 pm (shortest of all C-C bonds)
C≡C Bond Enthalpy	823 kJ/mol (strongest)
Bond composition	1 sigma + 2 pi bonds between the two carbons
First Stable Member	Ethyne (HC≡CH) also called Acetylene
Industrial Use	Oxyacetylene flame for welding ($C_2H_2 + O_2$)
Physical State	C_2-C_4 : gases C_5-C_{13} : liquids C_{14+} : solids

STRUCTURE OF TRIPLE BOND

Each carbon in alkyne: 2 sp hybridised orbitals + 2 unhybridised p orbitals

C-C sigma: head-on overlap of 2 sp orbitals

C-H sigma: sp orbital of C overlaps with 1s of H

2 C-C pi bonds: lateral overlap of 2p_x-2p_x and 2p_y-2p_y (perpendicular planes)

Electron cloud is cylindrically symmetrical --> Ethyne is a LINEAR molecule

Bond	Type	Enthalpy (kJ/mol)	Length (pm)
C-C in ethane	sigma only	348	154
C=C in ethene	sigma + pi	681	134
C≡C in ethyne	sigma + 2pi	823	120
C-C in benzene	delocalised	~507 (average)	139

PREPARATION OF ALKYNES

1. From Calcium Carbide (Industrial Method for Ethyne)



This is the industrial synthesis of acetylene

2. From Vicinal Dihalides (Dehydrohalogenation with alc. KOH)



Use alcoholic KOH (not aqueous) for elimination

CHEMICAL PROPERTIES OF ALKYNES

A. ACIDIC CHARACTER [NEET: High Frequency]

sp hybridised carbon has 50% s character --> highest electronegativity --> H more easily released as H⁺

Compound	C-H hybridisation	s character	Acid strength
Ethyne HC≡CH	sp	50%	MOST ACIDIC
Ethene CH ₂ =CH ₂	sp ²	33%	Less acidic
Ethane CH ₃ -CH ₃	sp ³	25%	LEAST ACIDIC

HC≡CH + Na --> HC≡C(-) Na(+) + 1/2 H₂ (Monosodium ethynide) CH₃-C≡C-H + NaNH₂ --> CH₃-C≡C(-) Na(+) + NH₃ (Propyne)

Acidic order: HC≡CH > CH₃-C≡CH >> CH₃-C≡C-CH₃ (internal — NOT acidic)

TRAP: Internal alkynes (but-2-yne) have NO terminal H on sp C --> do NOT react with Na or NaNH₂

B. ADDITION REACTIONS

Reaction	Reagent/Condition	Product	NEET Note
H ₂ (step 1)	Lindlar catalyst	cis-Alkene	Stops at alkene stage
H ₂ (step 1)	Na / liq. NH ₃	trans-Alkene	Birch reduction conditions
H ₂ (complete)	Pt/Pd/Ni	Alkane	Both pi bonds reduced
Br ₂	Br ₂ /CCl ₄	1st: dibromoalkene 2nd: tetrabromoalkane	Decolourises Br ₂ (unsaturation test)
HBr (1 eq.)	HBr (Markovnikov)	Vinyl bromide (bromoalkene)	Then gem-dihalide
HBr (2 eq.)	HBr (Markovnikov)	gem-Dihalide (2 X on same C)	gem = same carbon
H ₂ O	Hg ₂ +/H ₂ SO ₄ , 333K	Ethyne --> Ethanal Propyne --> Propanone	Via vinyl alcohol (tautomerism)

KEY TRAP — Hydration of Alkynes: HC≡CH + H₂O --> CH₂=CHOH (vinyl alcohol) --[tautomerism]--> CH₃CHO (Ethanal) CH₃-C≡CH + H₂O --> CH₃-CO-CH₃ (Propanone) -- Markovnikov rule

C. POLYMERISATION

Linear Polymerisation

n HC≡CH --> -(CH=CH-CH=CH)_n- (Polyacetylene) Conducts electricity under special conditions. Used in battery electrodes.

Cyclic Polymerisation [NEET: High Frequency]

3 HC≡CH --[red hot Fe tube, 873K]--> C₆H₆ (Benzene)

Best route from aliphatic to aromatic compounds

Memory: 3 acetylene molecules at 873K (red-hot iron) = benzene

AROMATIC HYDROCARBONS

Benzene and Its Derivatives — Arenes

BENZENE — QUICK FACTS

Property	Value
Molecular Formula	C ₆ H ₆
Hybridisation	sp ² (all 6 carbons)
C-C Bond Length	139 pm (between C-C 154 pm and C=C 133 pm)
Bond Angles	120° (regular hexagon, planar)
Isolated by	Michael Faraday, 1825
Structure proposed by	August Kekule, 1865
Physical state	Colourless liquid, characteristic pleasant aroma
Combustion	Burns with SOOTY/LUMINOUS flame (high C:H ratio = 1:1)
Reactivity	Prefers SUBSTITUTION over addition (to preserve aromaticity)

STRUCTURE OF BENZENE

Kekule Structure and Its Limitations

- Kekule proposed cyclic structure with alternating single and double bonds
- Predicted 2 ortho-disubstituted isomers — but only 1 is found experimentally
- Modified concept: oscillating double bonds — still failed to explain extra stability

Resonance and Stability

Benzene = resonance hybrid of two Kekule structures

Circle inside hexagon represents 6 DELOCALISED pi electrons

All 6 C-C bonds are equal = 139 pm (confirmed by X-ray diffraction)

Orbital Picture

Feature	Detail
All 6 C atoms	sp ² hybridised
Sigma framework	6 C-C sigma bonds + 6 C-H sigma bonds — all in one plane
Pi system	6 unhybridised p orbitals perpendicular to ring plane
Pi overlap	Forms 2 doughnut-shaped electron clouds (above and below the ring)
Resonance energy	~150 kJ/mol extra stability compared to hypothetical cyclohexatriene
Why substitution?	Addition would destroy aromaticity = loss of ~150 kJ/mol stability

WHY SUBSTITUTION NOT ADDITION? Addition destroys aromaticity (pi system) = loss of resonance energy (~150 kJ/mol). Substitution preserves aromaticity.

AROMATICITY — HUCKEL'S RULE [NEET: Very High Frequency]

A compound is **AROMATIC** if it has **ALL THREE** of the following:

- 1. PLANARITY** (flat ring)
- 2. Complete delocalisation of pi electrons in the ring**
- 3. $(4n + 2)$ pi electrons where $n = 0, 1, 2, 3...$ (Huckel's Rule)**

Compound	Pi electrons	n value	Aromatic?	Reason
Benzene	6	1	YES	$4(1)+2 = 6$
Naphthalene	10	2	YES	$4(2)+2 = 10$
Anthracene	14	3	YES	$4(3)+2 = 14$
Cyclopentadienyl anion ($C_5H_5^-$)	6	1	YES	6 pi electrons, planar
Cycloheptatrienyl cation ($C_7H_7^+$)	6	1	YES	6 pi electrons, planar
Cyclooctatetraene (COT)	8	—	NO	$4n$ pi (anti-aromatic candidate)
Cyclobutadiene	4	—	NO	$4n$ pi = ANTI-AROMATIC

ANTI-aromatic = $4n$ pi electrons (e.g. cyclobutadiene with 4 pi). DESTABILISED compound.

PREPARATION OF BENZENE

1. Cyclic polymerisation of Ethyne



2. Decarboxylation of Benzoic Acid



3. Reduction of Phenol



ELECTROPHILIC SUBSTITUTION REACTIONS [NEET: MOST IMPORTANT]

Reaction	Reagents/Conditions	Product	Electrophile
NITRATION	Conc. HNO_3 + Conc. H_2SO_4 323-333K (nitrating mixture)	Nitrobenzene $\text{C}_6\text{H}_5\text{NO}_2$	NO_2^+ (nitronium ion)
HALOGENATION	Cl_2 + Anhy. AlCl_3 (Lewis acid) Dark, room temp	Chlorobenzene $\text{C}_6\text{H}_5\text{Cl}$ + HCl	Cl^+ (chloronium ion)
SULPHONATION	Fuming H_2SO_4 (oleum) heat	Benzene sulphonic acid $\text{C}_6\text{H}_5\text{-SO}_3\text{H}$	SO_3 (electrophile)
FRIEDEL-CRAFTS ALKYLATION	R-Cl + Anhy. AlCl_3 Lewis acid catalyst	Alkylbenzene e.g. Toluene from CH_3Cl	R^+ (carbocation)
FRIEDEL-CRAFTS ACYLATION	RCOCl + Anhy. AlCl_3 or acid anhydride	Acylbenzene (aryl ketone) e.g. Acetophenone	RCO^+ (acylium ion)

EAS Mechanism — Step by Step

Step 1: Electrophile Generation

AlCl_3 (Lewis acid) activates reagent: $\text{Cl}_2 + \text{AlCl}_3 \rightarrow \text{Cl}^+ + \text{AlCl}_4^-$ (chloronium) $\text{CH}_3\text{Cl} + \text{AlCl}_3 \rightarrow \text{CH}_3^+ + \text{AlCl}_4^-$ (carbocation) $\text{HNO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{NO}_2^+ + \text{HSO}_4^-$ (nitronium — for nitration)

Step 2: Arenium Ion (sigma-complex) Formation

E⁺ attacks benzene ring --> one C becomes sp³ --> arenium ion (sigma-complex) formed Arenium ion stabilised by resonance (3 contributing structures) AROMATICITY IS LOST at this step

Step 3: Proton Removal — Aromaticity Restored

AlCl₄⁻ removes H⁺ from sp³ carbon --> AlCl₃ regenerated (catalyst recycled) AROMATICITY RESTORED --> substituted benzene formed

Addition Reactions (Special Conditions Only)

Benzene + 3H₂ --[Ni, heat, high P]--> Cyclohexane

Benzene + 3Cl₂ --[UV light, 500K]--> C₆H₆Cl₆ (BHC / Benzene hexachloride / Gammaxane)

BHC = ADDITION reaction (not substitution). UV light, NOT Lewis acid.

TRAP: Students write BHC as substitution reaction. It is ADDITION under UV light.

DIRECTIVE INFLUENCE [NEET: MOST REPEATED TOPIC]

Nature of substituent already on ring decides where next group enters

Group Type	Examples	Directs to	Effect on Ring	Key Point
ORTHO/PARA directors (Activating)	-OH, -NH ₂ , -NHR, -NR ₂ -OCH ₃ , -NHCOCH ₃ , -CH ₃	ortho and para positions	Activating (increases reactivity)	Lone pair donation (+M) increases e-density at o and p
ORTHO/PARA directors (Deactivating — TRAP!)	-F, -Cl, -Br, -I (all halogens)	ortho and para (NOT meta!)	DEACTIVATING (strong -I but +M)	MOST COMMON NEET TRAP Halogens = o/p despite deactivating
META directors (Deactivating)	-NO ₂ , -CN, -CHO, -COR -COOH, -COOR, -SO ₃ H	meta position	Strongly deactivating (-I and -M effects)	Withdraw e- from ring o,p positions electron-poor

MOST DANGEROUS NEET TRAP: Halogens (F, Cl, Br, I) are DEACTIVATING but ORTHO/PARA directors. Most students write meta = WRONG!

Logic for -OH (o/p): Lone pair on O donated to ring by resonance (+M effect) --> electron density increases at C1, C3 (ortho) and C5 (para)

Logic for -NO₂ (meta): Strong -I and -M effect --> withdraws electrons from ring --> ortho/para positions become electron-poor --> electrophile attacks less-poor meta position

CARCINOGENICITY AND TOXICITY

Polynuclear aromatic hydrocarbons (PAHs) with 2+ fused benzene rings are CARCINOGENIC

- 1,2-Benzanthracene
- 3-Methylcholanthrene
- 1,2-Benzpyrene
- 1,2,5,6-Dibenzanthracene
- 9,10-Dimethyl-1,2-benzanthracene

Formed during INCOMPLETE COMBUSTION of tobacco, coal, petroleum

Enter body --> biochemical reactions --> DAMAGE DNA --> cause cancer

COMPARISON TABLES

NEET Loves These — Master Every Comparison

1. ALKANE vs ALKENE vs ALKYNE

Property	Alkane	Alkene	Alkyne
Formula	C_nH_{2n+2}	C_nH_{2n}	C_nH_{2n-2}
Hybridisation	sp^3	sp^2	sp
Bond angle	109.5°	$\sim 120^\circ$	180°
C-C/C=C/C≡C bond length	154 pm	134 pm	120 pm
Bond enthalpy	348 kJ/mol	681 kJ/mol	823 kJ/mol
Geometry	Tetrahedral	Planar	Linear
Main reaction type	Free radical substitution	Electrophilic addition	Electrophilic addition
Acidity of C-H	Least acidic	Intermediate	Most acidic (terminal H)
Br ₂ /CCl ₄ test	Negative	Decolourises (positive)	Decolourises (positive)
Isomerism types	Chain + position	Chain + position + geometric	Chain + position only

2. MARKOVNIKOV vs ANTI-MARKOVNIKOV

Feature	Markovnikov	Anti-Markovnikov (Peroxide)
Applies to	HX (any) without peroxide	HBr ONLY with peroxide
Mechanism	Electrophilic addition (ionic)	Free radical chain
H adds to	Carbon with MORE H	Carbon with FEWER H
Br/X adds to	More substituted C (fewer H)	Less substituted C (more H)
For HBr + propene	2-Bromopropane (major)	1-Bromopropane (major)
Intermediate	2° carbocation (more stable)	2° free radical (more stable)
HCl/HI peroxide effect	NO peroxide effect	NO peroxide effect for HCl or HI
Discovered by	Markovnikov, 1869	Kharash and Mayo, 1933

3. cis vs trans ISOMERS

Property	cis Isomer	trans Isomer
Group arrangement	Same groups SAME side of C=C	Same groups OPPOSITE sides
Dipole moment	Higher (e.g. cis-but-2-ene = 0.33D)	Lower/zero (trans-but-2-ene ~ 0)
Boiling point	HIGHER (more polar)	LOWER
Melting point	LOWER	HIGHER (better crystal packing)
Stability	Less stable (steric strain)	More stable

Property	cis Isomer	trans Isomer
Example	cis-But-2-ene b.p. 277K	trans-But-2-ene b.p. 274K

4. ORTHO/PARA vs META DIRECTORS

Aspect	Ortho/Para Directors	Meta Directors
Effect on ring	Activating (except halogens)	Always deactivating
Electron effect	Increase e-density at o and p positions	Decrease e-density at o and p more than meta
Mechanism	+M or +I effect (electron donation)	-M and -I effects (electron withdrawal)
Examples	-OH, -NH ₂ , -OCH ₃ , -CH ₃ , -F, -Cl, -Br, -I	-NO ₂ , -CN, -CHO, -COOH, -SO ₃ H, -COOR
Halogen exception	Halogens = o/p BUT deactivating	—
Reactivity vs benzene	Faster (except halogens)	Slower
Product	Mainly ortho + para products	Mainly meta product

5. BOND LENGTH AND ENTHALPY ORDER

Bond	Length (pm)	Enthalpy (kJ/mol)	In compound
C-C (single, sp ³ -sp ³)	154	348	Alkanes
C=C (double)	134 (133)	681	Alkenes
C≡C (triple)	120	823	Alkynes
C-C in benzene	139	~507 (delocalised average)	Benzene (arenes)
C-H (sp ³)	112	~414	Alkanes
C-H (sp ²)	~108	~444	Alkenes
C-H (sp)	~106	~506	Alkynes

Trend: More s-character --> shorter bond + stronger bond + more acidic C-H

NEET PYQ PATTERN ANALYSIS

What Gets Asked — Ranked and Analysed

TIER 1 — MUST MEMORIZE (Repeated 3+ Times in NEET/AIPMT)

Concept	Times	Key Fact to Remember
Markovnikov Rule + Product	5+	2° carbocation more stable --> Markovnikov product
Anti-Markovnikov (Peroxide)	4+	ONLY HBr shows peroxide effect; gives 1-bromopropane
Directive influence (o/m/p)	5+	Halogens = o/p (TRAP!); -NO ₂ = meta
Ozonolysis products	4+	Each C of C=C becomes carbonyl
Lindlar vs Na/liq.NH ₃	4+	Lindlar = CIS; Na/liq.NH ₃ = TRANS
Aromaticity Huckel Rule	4+	(4n+2)pi = aromatic; 4n pi = anti-aromatic
Free radical mechanism	3+	Ethane byproduct from termination CH ₃ . + CH ₃ .
Acidic character of alkynes	3+	Terminal sp C-H most acidic; internal alkynes not acidic
Geometrical isomerism	3+	Both C of C=C must have 2 DIFFERENT groups
Kolbe electrolysis	3+	Methane cannot be made; gives even-carbon alkanes

TIER 2 — HIGH PROBABILITY (Repeated 2 Times)

Concept	Key Point	Common Q Format
Wurtz reaction	Only even-carbon, same halide	Identify product or limitation
Decarboxylation	Product has 1 less C than acid	What acid gives propane?
Conformations of ethane	Staggered most stable; diff = 12.5 kJ/mol	Which is more stable?
Hydration of alkynes	Ethyne --> ethanal; propyne --> propanone	Identify carbonyl product
Baeyer's reagent test	Cold dil. KMnO ₄ --> vicinal diol	Test for unsaturation
3 HC≡CH --> benzene	Red-hot Fe tube, 873K	Temperature/condition Q
BHC (Gammexane)	UV + 3Cl ₂ ; ADDITION not substitution	Mechanism type Q
Resonance stability of benzene	~150 kJ/mol extra stability	Why substitution preferred?

TIER 3 — TRAP ZONE (Common Wrong Answers)

Trap Question	WRONG Answer (students pick)	CORRECT Answer
Halogens direct to?	Meta	Ortho/Para
Peroxide + HCl =?	Anti-Markovnikov	Markovnikov (no peroxide effect for HCl)
cis- has higher m.p. or b.p.?	cis higher m.p.	cis higher b.p.; trans higher m.p.
Methane via Kolbe?	Yes, possible	NOT possible
Staggered stability	Eclipsed more stable	Staggered more stable

Trap Question	WRONG Answer (students pick)	CORRECT Answer
Alkyne + H ₂ O product	Alcohol	Carbonyl compound (tautomerism)
But-2-yne + NaNH ₂	Reacts (acidic)	No reaction (internal alkyne)
C-C length in benzene	134 pm or 154 pm	139 pm (resonance hybrid)
BHC formation	Substitution reaction	Addition reaction under UV light

FORMULA AND EXCEPTION SHEET

One-Look Rapid Revision Reference

GENERAL FORMULAS

Series	Formula	First Member
Alkanes	C_nH_{2n+2}	CH ₄ (Methane)
Cycloalkanes	C_nH_{2n}	C ₃ H ₆ (Cyclopropane)
Alkenes	C_nH_{2n}	C ₂ H ₄ (Ethene)
Alkynes	C_nH_{2n-2}	C ₂ H ₂ (Ethyne)
Benzene homologues	C_nH_{2n-6}	C ₆ H ₆ (Benzene)
Alkyl groups	C_nH_{2n+1}	CH ₃ - (Methyl)

REACTIONS — REAGENTS AND CONDITIONS QUICK TABLE

Reaction	Reagent	Condition	Product
Hydrogenation	Pt/Pd	Room temp	Alkane from alkene/alkyne
Hydrogenation (Ni)	Ni	High T and P	Alkane
Cis-alkene from alkyne	H ₂ + Lindlar catalyst (Pd/C + quinoline)	Room temp	cis-Alkene
Trans-alkene from alkyne	H ₂ + Na / liq. NH ₃	Low temp	trans-Alkene
Dehydrohalogenation	Alc. KOH, heat	Beta-elimination	Alkene from alkyl halide
Dehalogenation	Zn	Heat	Alkene from vicinal dihalide
Dehydration of alcohol	Conc. H ₂ SO ₄ , 170C	Beta-elimination	Alkene
Halogenation of alkene	X ₂ / CCl ₄	Dark, room temp	Vicinal dihalide
HBr + alkene (Markovnikov)	HBr, no peroxide	Dark	2° alkyl bromide
HBr + alkene (Anti-Markovnikov)	HBr + peroxide (ROOR)	Peroxide	1° alkyl bromide
Hydration of alkene	H ₂ O + dil. H ₂ SO ₄	Markovnikov	Alcohol
Baeyer's test (oxidation)	Cold dil. KMnO ₄	273K, alkaline	Vicinal diol
Ozonolysis (reductive)	O ₃ , then Zn/H ₂ O	Reductive	Aldehydes/Ketones
Ozonolysis (oxidative)	O ₃ , then H ₂ O ₂	Oxidative	Carboxylic acids
Hydration of alkynes	H ₂ O + Hg ²⁺ /H ₂ SO ₄	333K	Aldehyde (terminal) or ketone
Cyclic polymerisation	Red hot Fe tube	873K	Benzene from 3 acetylene
Aromatization (alkanes)	Cr ₂ O ₃ /V ₂ O ₅ on Al ₂ O ₃	773K, 10-20 atm	Benzene from n-hexane
Decarboxylation	NaOH + CaO (soda lime)	Heat, fusion	Alkane (1 C less than acid)
Wurtz reaction	2Na, dry ether	R-X + R-X	Higher alkane (R-R)

Reaction	Reagent	Condition	Product
Kolbe electrolysis	Electrolysis (aqueous)	Anode reaction	Even-carbon alkane
Nitration of benzene	Conc. HNO ₃ + Conc. H ₂ SO ₄	323-333K	Nitrobenzene
Halogenation of benzene	X ₂ + Anhy. AlCl ₃ /FeCl ₃	Dark, Lewis acid	Halobenzene + HX
Sulphonation of benzene	Fuming H ₂ SO ₄ (Oleum)	Heat, reversible	Benzene sulphonic acid
Friedel-Crafts alkylation	RCI + Anhy. AlCl ₃	Lewis acid	Alkylbenzene
Friedel-Crafts acylation	RCOCl + Anhy. AlCl ₃	Lewis acid	Acybenzene (ketone)
Reduction of benzene	3H ₂ + Ni	High T and P	Cyclohexane
BHC formation	3Cl ₂ + UV light	500K	C ₆ H ₆ Cl ₆ (BHC/Gammaxane)

EXCEPTION AND SPECIAL CASES

Exception	Explanation
Methane CANNOT be made by Kolbe method	Formate gives only CO ₂ , no CH ₃ radical
Iodination of alkanes needs HIO ₃ or HNO ₃	I ₂ reaction is reversible; oxidising agent shifts equilibrium
Fluorination is uncontrollable	F-F bond too reactive; complete degradation
ONLY HBr shows anti-Markovnikov (peroxide effect)	F ₂ too strong, H-I too weak for radical chain
Lindlar catalyst = Pd/C + quinoline or sulphur	Partially deactivated -- stops at alkene stage
Internal alkynes do NOT react with NaNH ₂	Only terminal sp C-H is acidic
Benzene shows substitution not addition	Addition destroys aromaticity = loss of ~150 kJ/mol
Halogens are o/p directors despite deactivating	+M effect (lone pair) overrides -I for directing
BHC = addition not substitution	UV light + Cl ₂ ; free radical; no Lewis acid
trans isomer has higher melting point	More symmetrical --> better crystal packing
Wurtz with 2 different R-X gives mixture	R-R, R-R' and R'-R' all form (not selective)
C-C bond in benzene = 139 pm (not 133 or 154)	Resonance hybrid -- intermediate value

MEMORY TRICKS AND MNEMONICS

Never Forget — Smart Hooks for Weak Students

MARKOVNIKOV — "RICH GET RICHER"

H adds to carbon that ALREADY HAS MORE H atoms (the rich carbon gets richer) Br adds to the poor carbon (fewer H, more substituted)

ANTI-MARKOVNIKOV — "ONLY BROMINE REBELS"

Anti-Markovnikov = Only HBr + Peroxide NOT HCl (bond too strong), NOT HI (I radical recombines) Memory: Bromine is the only rebel halide in radical reactions

LINDLAR vs Na/NH₃ — "L FOR LEFT = SAME SIDE = CIS"

Lindlar = CIS alkene (same side = L shape = cis) Na/liq.NH₃ = TRANS alkene (opposite sides) Hook: Lindlar = Like = same = cis

ORTHO/PARA DIRECTORS — "OH AMiNe Gets Pretty Fast"

-OH, -NH₂ (amine), -NHR, -NR₂, -OCH₃, Alkyl (-CH₃, -C₂H₅) All donate electrons by resonance (+M) --> o/p directors

META DIRECTORS — "NO CN CHO COR COOH SO₃H"

-NO₂, -CN, -CHO, -COR, -COOH, -COOR, -SO₃H All withdraw electrons from ring --> meta directors Hook: META directors = electron THIEVES

HUCKEL RULE — "2, 6, 10, 14 = MAGIC AROMATIC NUMBERS"

n=0 --> 2 pi electrons | n=1 --> 6 pi electrons (benzene) n=2 --> 10 pi electrons (naphthalene) | n=3 --> 14 pi Quick check: Is the number 2, 6, 10, 14... ? Then aromatic.

STAGGERED vs ECLIPSED — "STAY STAGGERED = STAY STABLE"

Staggered = S for Safe and Stable (most stable, least strain) Eclipsed = E for Exhausted and Energetic (highest energy, least stable) Energy diff = 12.5 kJ/mol (very small, rotation practically free)

OZONOLYSIS SHORTCUT — "BREAK THE BOND, BOTH SIDES GET =O"

Each C of C=C becomes carbonyl carbon after ozonolysis: Terminal =CH₂ --> HCHO (methanal/formaldehyde) =CHR --> RCHO (aldehyde) =CR₂ --> R₂C=O (ketone)

ACIDITY ORDER — "SP ALWAYS WINS"

sp (alkyne) > sp² (alkene) > sp³ (alkane) in acidity More s-character = more electronegative C = weaker C-H = more acidic

HALOGENATION RATE — "F Cl Br I = Fast Cat But Idle"

F₂ > Cl₂ > Br₂ > I₂ (decreasing reactivity with alkanes) F = Ferocious (too fast), I = Idle (too slow, reversible)

REACTIVITY ORDER QUICK CARDS

Category	Order (High to Low)	Memory Hook
Halogenation of alkanes	$F_2 > Cl_2 > Br_2 > I_2$	F Calls Busy Ivan
H replacement rate (alkanes)	$3^\circ > 2^\circ > 1^\circ$	Tertiary = Toughest to stay
HX addition rate	$HI > HBr > HCl$	I Beat Chlorine
C-H acidity	$sp > sp^2 > sp^3$ (alkyne > alkene > alkane)	sp wins always
C-H bond strength	$sp > sp^2 > sp^3$	More s character = stronger
C-H bond length	$sp^3 > sp^2 > sp$	More s character = shorter
Boiling point (alkane isomers)	Straight chain > branched chain	Straight is hotter

RAPID REVISION SHEETS

5-Minute Review | Last Night Before NEET

5-MINUTE RAPID REVISION — KEY POINTS ONLY

- C_nH_{2n+2} = Alkanes (sp^3 , paraffins, free radical substitution)
- C_nH_{2n} = Alkenes (sp^2 , olefins, electrophilic addition)
- C_nH_{2n-2} = Alkynes (sp , linear, electrophilic addition, acidic terminal H)
- Markovnikov: H^+ to C with MORE H; forms more stable carbocation
- Anti-Markovnikov: ONLY HBr + peroxide; free radical mechanism
- Lindlar $\rightarrow$ CIS alkene | Na/liq.NH₃ $\rightarrow$ TRANS alkene
- Ozonolysis: $C=C$ breaks $\rightarrow$ each C becomes $C=O$ group
- Decarboxylation: $RCOONa + NaOH/CaO \rightarrow R-H$ (product has 1 C less)
- Wurtz: $2RX + 2Na$ (dry ether) $\rightarrow R-R$ | Only EVEN carbon, SAME halide
- Kolbe: electrolysis of $RCOO^-Na^+$ $\rightarrow R-R$ at anode (even C) | NO methane by Kolbe
- Halogenation rate: $F_2 > Cl_2 > Br_2 > I_2$ | H reactivity: $3^\circ > 2^\circ > 1^\circ$
- $CH_4 + Cl_2$ (UV) $\rightarrow CH_3Cl \rightarrow CH_2Cl_2 \rightarrow CHCl_3 \rightarrow CCl_4$ (stepwise)
- Ethane = BYPRODUCT of methane chlorination (termination: $CH_3\cdot + CH_3\cdot \rightarrow C_2H_6$)
- Benzene: 6 π e⁻, sp^2 , 139 pm C-C, resonance hybrid, prefers EAS
- Aromaticity = planar + $(4n+2)$ π electrons + delocalised
- Ortho/Para directors: -OH, -NH₂, -OCH₃, -CH₃, -F, -Cl, -Br (halogens = deactivating but o/p!)
- Meta directors: -NO₂, -CN, -CHO, -COOH (all electron-withdrawing)
- EAS mechanism: E^+ generated $\rightarrow$ arenium ion (sp^3) $\rightarrow H^+$ removed $\rightarrow$ aromaticity restored
- BHC = Benzene + $3Cl_2$ under UV (ADDITION not substitution)
- Cyclohexane = benzene + $3H_2$ under Ni (addition under special conditions)
- Staggered conformation = MORE stable (min torsional strain, diff = 12.5 kJ/mol)
- cis: same side, higher b.p., lower m.p. | trans: opposite, lower b.p., higher m.p.
- Acidic order: $HC=CH > CH_2=CH_2 > CH_3-CH_3$ ($sp > sp^2 > sp^3$)
- ONLY terminal alkynes react with Na or NaNH₂ (internal alkynes do not)
- Hydration of alkynes: Hg^{2+}/H_2SO_4 , 333K $\rightarrow$ carbonyl compound (Markovnikov)
- $3 HC=CH \rightarrow$ benzene (red-hot Fe tube, 873K) = cyclic polymerisation

MOST FORGETTABLE — READ NIGHT BEFORE EXAM

Methane CANNOT be prepared by Kolbe electrolysis

Peroxide effect: ONLY HBr, NOT HCl or HI

Halogens = ortho/para directors (deactivating but o/p!)

BHC = addition reaction of benzene + $3Cl_2$ under UV light

Lindlar catalyst $\rightarrow$ CIS alkene; Na/liq.NH₃ $\rightarrow$ TRANS alkene

C-C bond in benzene = 139 pm (not 133, not 154)

Staggered = most stable (NOT eclipsed)

trans isomer has HIGHER melting point than cis

Internal alkynes are NOT acidic (no terminal sp C-H)

Ethane is BYPRODUCT during chlorination of methane

$(4n+2)$ pi for aromaticity; cyclobutadiene (4 pi) = ANTI-aromatic

Hydration of alkynes gives CARBONYL compound (via vinyl alcohol tautomerism)

NEET-LEVEL MCQ TEST

30 Questions — Fully Solved with Trap Explanations

1. Addition of HBr to propene in presence of benzoyl peroxide gives:

- A) 2-Bromopropane
- B) 1-Bromopropane
- C) Allyl bromide
- D) 1,2-Dibromopropane

Answer: B) 1-Bromopropane | Topic: Anti-Markovnikov

Explanation: Peroxide effect (anti-Markovnikov) with HBr only. Br radical adds to terminal C (less substituted). 2° radical more stable than 1° radical.

2. Which compound will NOT show cis-trans isomerism?

- A) $\text{CH}_3\text{-CH=CH-CH}_3$
- B) $\text{CH}_3\text{-CH=CCl-CH}_3$
- C) $(\text{CH}_3)_2\text{C=CH-C}_2\text{H}_5$
- D) $\text{CH}_3\text{CH=C(CH}_3)_2$

Answer: C) $(\text{CH}_3)_2\text{C=CH-C}_2\text{H}_5$ and D) $\text{CH}_3\text{CH=C(CH}_3)_2$ | Topic: Geometrical Isomerism

Explanation: For cis-trans isomerism, BOTH doubly bonded carbons need 2 DIFFERENT groups. In (C) left C has 2 CH_3 (identical). In (D) right C has 2 CH_3 (identical). So C and D do not show geometric isomerism. NEET usually gives only one option — pick the one where one C has 2 identical groups.

3. Ozonolysis of 2-methylbut-2-ene gives:

- A) Ethanal + Propanal
- B) Propanone + Ethanal
- C) Butanal + Methanal
- D) Propanone + Formaldehyde

Answer: B) Propanone + Ethanal (same as D in IUPAC names) | Topic: Ozonolysis

Explanation: 2-Methylbut-2-ene = $(\text{CH}_3)_2\text{C=CH-CH}_3$. Cleavage at C=C : $(\text{CH}_3)_2\text{C=}$ gives propanone. $=\text{CH-CH}_3$ gives ethanal.

4. Which catalyst gives trans-alkene from an alkyne?

- A) Lindlar catalyst
- B) Pt/C
- C) Na/liquid ammonia
- D) $\text{Ni at } 200^\circ\text{C}$

Answer: C) Na / liquid ammonia | Topic: Lindlar vs Na/NH_3

Explanation: Lindlar = CIS alkene (syn addition). Na/liquid NH_3 = TRANS alkene (anti addition). Most frequently tested NEET question.

5. Ethane is obtained as a byproduct during chlorination of methane. This is because of:

- A) Initiation step
- B) Propagation step (a)
- C) Propagation step (b)
- D) Termination step: $\text{CH}_3 \cdot + \text{CH}_3 \cdot \rightarrow \text{C}_2\text{H}_6$

Answer: D) Termination step: $\text{CH}_3 \cdot + \text{CH}_3 \cdot \rightarrow \text{C}_2\text{H}_6$ | Topic: Free Radical Mechanism

Explanation: In termination step, two methyl radicals combine to give ethane. This explains ethane as byproduct in methane chlorination.

6. Which of the following is an ortho and para director?

- A) $-\text{NO}_2$

- B) -CN
- C) -Cl
- D) -COOH

Answer: C) -Cl | Topic: Directive Influence — TRAP

Explanation: Halogens (-Cl, -Br, -F, -I) are ortho/para directors despite being deactivating. This is the most common trap in NEET. -NO₂, -CN, -COOH are all meta directors.

7. Aromaticity requires:

- A) 4n pi electrons
- B) (4n+2) pi electrons, planar, delocalised
- C) Only benzene ring present
- D) Alternating single and double bonds

Answer: B) (4n+2) pi electrons, planar, delocalised | Topic: Huckel Rule

Explanation: Huckel rule: aromatic = (4n+2) pi electrons + planar + complete delocalisation. 4n = anti-aromatic. Non-benzenoid aromatics also exist.

8. Product of hydration of propyne with Hg²⁺/H₂SO₄ at 333K is:

- A) Propanal
- B) Propan-1-ol
- C) Propanone
- D) Propan-2-ol

Answer: C) Propanone | Topic: Hydration of Alkynes

Explanation: CH₃-C≡CH + H₂O → CH₃-C(OH)=CH₂ (vinyl alcohol) → [tautomerism] → CH₃-CO-CH₃ (propanone). Markovnikov: OH adds to middle C of propyne.

9. Wurtz reaction is NOT suitable for preparing:

- A) n-Butane from bromoethane
- B) n-Hexane from 1-bromopropane
- C) Ethane from bromomethane
- D) n-Propane from bromomethane + bromoethane

Answer: D) n-Propane from bromomethane + bromoethane | Topic: Wurtz Limitation

Explanation: Two different alkyl halides give mixture of R-R, R-R' and R'-R'. Not selective for odd carbon or mixed halide reactions.

10. The most stable conformation of ethane is:

- A) Eclipsed
- B) Skew
- C) Staggered
- D) Gauche

Answer: C) Staggered | Topic: Conformations

Explanation: Staggered: H atoms farthest apart, minimum electron cloud repulsion, minimum torsional strain. 12.5 kJ/mol less energy than eclipsed.

11. C-C bond length in benzene (139 pm) is:

- A) Equal to C-C in ethane (154 pm)
- B) Equal to C=C in ethene (134 pm)
- C) Intermediate between C-C and C=C
- D) Equal to C≡C in ethyne (120 pm)

Answer: C) Intermediate between C-C and C=C | Topic: Benzene Structure

Explanation: Benzene = resonance hybrid. C-C = 139 pm, between C-C (154) and C=C (133). All 6 C-C bonds are equal.

12. Which of the following shows peroxide effect?

- A) HCl + propene + ROOR

- B) HBr + but-1-ene + ROOR
- C) HI + ethene + ROOR
- D) Br₂ + ethene + ROOR

Answer: B) HBr + but-1-ene + ROOR | Topic: Peroxide Effect — Trap

Explanation: Peroxide effect only for HBr. H-Cl bond too strong (430 kJ/mol). I radicals combine to I₂ instead of adding.

13. Baeyer's reagent is:

- A) Acidic KMnO₄
- B) Cold dilute alkaline KMnO₄
- C) K₂Cr₂O₇ in H₂SO₄
- D) Hot KMnO₄

Answer: B) Cold dilute alkaline KMnO₄ | Topic: Baeyer's Reagent

Explanation: Baeyer reagent = cold dilute alkaline KMnO₄. Alkene → vicinal diol. Pink colour decolourises = positive test for unsaturation.

14. Which compound undergoes nitration most easily?

- A) Benzene
- B) Toluene
- C) Nitrobenzene
- D) Chlorobenzene

Answer: B) Toluene | Topic: EAS Reactivity

Explanation: Toluene: methyl group = +I effect, activating, o/p director → fastest reaction. Reactivity: Toluene > Benzene > Chlorobenzene > Nitrobenzene.

15. Which statement about Kolbe electrolysis is CORRECT?

- A) Gives odd-carbon alkanes
- B) Gives even-carbon alkanes at anode
- C) Methane can be prepared
- D) Uses non-aqueous solution

Answer: B) Gives even-carbon alkanes at anode | Topic: Kolbe Method

Explanation: 2RCOO⁻ → R-R + 2CO₂ at anode. Each R combines → even-carbon alkane. Methane cannot be made. Aqueous solution used.

16. Which compound is NOT aromatic?

- A) Benzene (6 π)
- B) Cyclopentadienyl anion (6 π)
- C) Cycloheptatrienyl cation (6 π)
- D) Cyclooctatetraene (8 π)

Answer: D) Cyclooctatetraene (8 π) | Topic: Huckel Rule Application

Explanation: COT has 8 π = 4n (anti-aromatic candidate), also tub-shaped (non-planar). Benzene, Cp anion, tropylium cation all have 6 π = aromatic.

17. Which reagent converts but-1-yne to cis-but-1-ene?

- A) Pt/C + H₂ (excess)
- B) Pd/C + H₂ (excess)
- C) Lindlar catalyst + H₂
- D) Na + liq. NH₃

Answer: C) Lindlar catalyst + H₂ | Topic: Partial Hydrogenation

Explanation: Lindlar = Pd/C partially deactivated with quinoline/sulphur. Adds H₂ and stops at alkene. Gives CIS alkene (syn addition).

18. Sodium metal reacts with ethyne to produce:

- A) Sodium ethanoate + H₂

- B) Sodium acetylide + H₂
- C) Sodium ethoxide + H₂
- D) No reaction

Answer: B) Sodium acetylide + H₂ | Topic: Acidic Alkynes

Explanation: $\text{HC}\equiv\text{CH} + \text{Na} \rightarrow \text{HC}\equiv\text{C-Na}^+ + 1/2 \text{H}_2$. Terminal sp C-H is acidic (50% s-character). This does not happen with ethane or ethene.

19. Decarboxylation of sodium propanoate gives:

- A) Propane
- B) Ethane
- C) Methane
- D) Butane

Answer: B) Ethane | Topic: Decarboxylation Trap

Explanation: $\text{CH}_3\text{CH}_2\text{COONa} + \text{NaOH} \xrightarrow{[\text{CaO}]} \text{CH}_3\text{CH}_3 + \text{Na}_2\text{CO}_3$. Product has 1 C less than acid. Propanoic acid (3C) $\rightarrow$ ethane (2C). TRAP: students often say propane.

20. Benzene + Cl₂ + anhydrous AlCl₃ (dark) gives:

- A) Cyclohexyl chloride
- B) Chlorobenzene (EAS)
- C) BHC / Gammaxane
- D) Benzaldehyde

Answer: B) Chlorobenzene (electrophilic substitution) | Topic: EAS vs Free Radical — Trap

Explanation: Anhydrous AlCl₃ (Lewis acid) in dark = EAS $\rightarrow$ Chlorobenzene + HCl. BHC requires UV light + 3Cl₂ (free radical addition). Critical distinction.

21. Assertion: trans-but-2-ene has higher melting point than cis-but-2-ene. Reason: trans isomer is more symmetrical and packs better in crystal lattice.

- A) Both A and R true; R is correct explanation
- B) Both true; R not correct explanation
- C) A true but R false
- D) A false but R true

Answer: A) Both A and R are true; R is the correct explanation | Topic: cis-trans Properties (AR type)

Explanation: trans isomers are more symmetrical $\rightarrow$ better crystal packing $\rightarrow$ higher lattice energy $\rightarrow$ higher m.p. cis has higher b.p. (more polar).

22. Hybridisation and geometry of carbon atoms in ethyne are:

- A) sp³, tetrahedral
- B) sp², planar
- C) sp, linear
- D) sp, bent

Answer: C) sp, linear | Topic: Hybridisation of Alkynes

Explanation: Each C in ethyne = sp hybridised. sp $\rightarrow$ linear geometry. H-C $\equiv$ C-H bond angle = 180°. Ethyne is a linear molecule.

23. In Friedel-Crafts acylation, the electrophile is:

- A) AlCl₄⁻
- B) Acylium ion (RCO⁺)
- C) Chlorine radical
- D) AlCl₃

Answer: B) Acylium ion (RCO⁺) | Topic: Friedel-Crafts Acylation

Explanation: $\text{RCOCl} + \text{AlCl}_3 \rightarrow \text{RCO}^+ + \text{AlCl}_4^-$. Acylium = electrophile. No rearrangement (unlike alkylation) because acylium is resonance-stabilised.

24. Which alkyne does NOT react with NaNH₂?

- A) Ethyne ($\text{HC}=\text{CH}$)
- B) Propyne ($\text{CH}_3\text{C}=\text{CH}$)
- C) But-1-yne
- D) But-2-yne ($\text{CH}_3\text{-C}=\text{C-CH}_3$)

Answer: D) But-2-yne ($\text{CH}_3\text{-C}=\text{C-CH}_3$) | Topic: Acidic Alkynes — Trap

Explanation: NaNH_2 reacts with TERMINAL alkynes only. But-2-yne is internal (no H on triply bonded C) $\rightarrow$ no acidic H $\rightarrow$ no reaction.

25. In EAS mechanism, the arenium ion (sigma-complex) is formed when:

- A) E^+ attacks outside the ring
- B) E^+ attacks pi-cloud, one C becomes sp^3
- C) Proton is removed from ring
- D) AlCl_4^- attacks ring

Answer: B) E^+ attacks pi-cloud, one C becomes sp^3 | Topic: EAS Mechanism

Explanation: Electrophile (E^+) attacks electron-rich benzene pi cloud $\rightarrow$ one C becomes sp^3 $\rightarrow$ sigma-complex (arenium ion) forms. Aromaticity lost. Then H^+ removed to restore aromaticity.

26. The number of sigma and pi bonds in ethyne ($\text{HC}=\text{CH}$) are respectively:

- A) 2 sigma, 1 pi
- B) 3 sigma, 2 pi
- C) 2 sigma, 3 pi
- D) 3 sigma, 1 pi

Answer: B) 3 sigma, 2 pi | Topic: Bond Counting

Explanation: $\text{HC}=\text{CH}$: C-H (sigma) + C-H (sigma) + C-C (sigma) = 3 sigma. The triple bond = 1 sigma + 2 pi. Total: 3 sigma + 2 pi.

27. Which of the following gives formaldehyde (HCHO) on ozonolysis?

- A) But-2-ene
- B) 2-Methylbut-2-ene
- C) Propene
- D) But-2-ene (internal only)

Answer: C) Propene | Topic: Ozonolysis Shortcut

Explanation: Propene = $\text{CH}_3\text{-CH}=\text{CH}_2$. Terminal $=\text{CH}_2 \rightarrow \text{HCHO}$ (formaldehyde). $\text{CH}_3\text{-CH=}$ $\rightarrow$ CH_3CHO (ethanal). But-2-ene ($\text{CH}_3\text{-CH}=\text{CH-CH}_3$) gives 2 x ethanal; no HCHO .

28. Which is a correct statement about Wurtz reaction?

- A) Gives odd-carbon alkanes
- B) Two different halides always give one pure product
- C) Gives even-carbon alkanes when same halide used
- D) Uses aqueous sodium

Answer: C) Gives even-carbon alkanes when same halide used | Topic: Wurtz Reaction

Explanation: $2\text{RX} + 2\text{Na} \rightarrow \text{R-R}$. Same halide = one product. Different halides = mixture. Dry ether solvent. Even carbon alkanes formed.

29. Which statement about benzene is INCORRECT?

- A) All C-C bonds equal (139 pm)
- B) Planar molecule
- C) Preferentially undergoes addition reactions
- D) Has 6 delocalised pi electrons

Answer: C) Preferentially undergoes addition reactions | Topic: Benzene Properties

Explanation: Benzene preferentially undergoes SUBSTITUTION (not addition). Addition destroys aromaticity = loss of ~ 150 kJ/mol resonance energy.

30. Arrange ethyne, benzene, n-hexane in DECREASING order of acidic behaviour:

- A) n-Hexane > Benzene > Ethyne

- B) Benzene > Ethyne > n-Hexane
- C) Ethyne > Benzene > n-Hexane
- D) Ethyne > n-Hexane > Benzene

Answer: C) Ethyne > Benzene > n-Hexane | Topic: Acidity Order (NCERT Question)

Explanation: Ethyne: sp C-H (50% s-character, most acidic). Benzene: sp^2 C-H (33% s-character). n-Hexane: sp^3 C-H (25% s-character, least acidic). NEET direct question from NCERT.

GOLDEN SECTION

Top 20 of Everything — Ultimate Final Revision

TOP 20 REACTIONS FOR NEET

#	Reaction / Concept	Frequency
1	HBr + propene (Markovnikov) --> 2-Bromopropane	Very High
2	HBr + propene + ROOR (Anti-Markovnikov) --> 1-Bromopropane	Very High
3	Alkyne + H ₂ /Lindlar --> cis-alkene + Na/liq.NH ₃ --> trans-alkene	Very High
4	Ozonolysis: C=C + O ₃ then Zn/H ₂ O --> aldehydes/ketones	Very High
5	Benzene + HNO ₃ /H ₂ SO ₄ --> Nitrobenzene (nitration)	Very High
6	Benzene + Cl ₂ /AlCl ₃ --> Chlorobenzene (halogenation)	Very High
7	Benzene + RCl/AlCl ₃ --> Alkylbenzene (Friedel-Crafts alkylation)	Very High
8	Benzene + 3Cl ₂ /UV --> BHC/Gammaxane (free radical addition)	High
9	3 HC=CH --> Benzene (red-hot Fe, 873K — cyclic polymerisation)	High
10	RCOONa + NaOH/CaO --> R-H (decarboxylation — 1C less)	High
11	2RX + 2Na (dry ether) --> R-R (Wurtz — EVEN carbon only)	High
12	HC=CH + H ₂ O (Hg ²⁺ /H ⁺ , 333K) --> CH ₃ CHO (Markovnikov hydration)	High
13	CH ₃ -C=CH + H ₂ O --> CH ₃ -CO-CH ₃ (propanone, Markovnikov)	High
14	Alkene + cold dil. KMnO ₄ (Baeyer) --> diol (decolourises)	High
15	n-Hexane --> Benzene (Cr ₂ O ₃ , 773K, 10-20 atm — aromatization)	Medium
16	Alkyl halide + alc. KOH --> alkene (dehydrohalogenation)	Medium
17	Benzene + 3H ₂ /Ni --> Cyclohexane (addition under special conditions)	Medium
18	Kolbe: RCOO-Na ⁺ (electrolysis) --> R-R at anode (EVEN carbon)	Medium
19	Benzene + fuming H ₂ SO ₄ --> Benzene sulphonic acid (sulphonation)	Medium
20	Benzene + CH ₃ COCl/AlCl ₃ --> Acetophenone (Friedel-Crafts acylation)	Medium

TOP 20 ONE-LINERS

- 1. Alkanes = C_nH_{2n+2} | sp³ | 109.5° | 154 pm C-C | paraffins (inert)
- 2. Alkenes = C_nH_{2n} | sp² | ~120° | 134 pm C=C | olefins (oil forming)
- 3. Alkynes = C_nH_{2n-2} | sp | 180° | 120 pm C≡C | linear | most acidic C-H
- 4. Markovnikov: H⁺ to C with MORE H atoms (forms more stable carbocation)
- 5. Anti-Markovnikov (peroxide): ONLY HBr | free radical mechanism | 1-bromopropane from propene
- 6. Lindlar = CIS alkene | Na/liq.NH₃ = TRANS alkene (from partial hydrogenation of alkynes)
- 7. Ozonolysis: each C of C=C becomes C=O | terminal =CH₂ gives HCHO
- 8. Huckel Rule: (4n+2) pi electrons + planar + delocalised = AROMATIC
- 9. Benzene: 139 pm C-C | sp² | 6 pi electrons (delocalised) | prefers EAS (not addition)
- 10. Halogens = DEACTIVATING but ORTHO/PARA directors — NEET TRAP!

- 11. $-\text{NO}_2$, $-\text{COOH}$, $-\text{CN}$ = META directors + deactivating
- 12. $-\text{OH}$, $-\text{NH}_2$, $-\text{OCH}_3$ = ACTIVATING + ORTHO/PARA directors
- 13. BHC = Benzene + 3Cl_2 (UV, 500K) — ADDITION not substitution
- 14. Staggered more stable than eclipsed | energy diff = 12.5 kJ/mol
- 15. cis: higher b.p., lower m.p. | trans: lower b.p., higher m.p.
- 16. Internal alkynes NOT acidic (no terminal H on sp C)
- 17. Methane CANNOT be prepared by Kolbe electrolysis
- 18. Ethane = BYPRODUCT of methane chlorination (termination: $\text{CH}_3\cdot + \text{CH}_3\cdot$)
- 19. Wurtz: useful only for EVEN carbon, SAME alkyl halide
- 20. Acidic order: Alkyne > Alkene > Alkane (sp > sp² > sp³ C-H)

TOP 20 MISTAKES STUDENTS MAKE

#	Mistake	Correct Fact
1	Halogens = meta directors	Halogens = ortho/para directors (deactivating but o/p)
2	Anti-Markovnikov with HCl	Only HBr shows peroxide effect
3	Methane by Kolbe possible	Formate $\rightarrow$ CO ₂ only; methane impossible
4	Staggered = less stable	Staggered = MOST stable (least torsional strain)
5	trans has higher boiling point	trans has higher m.p.; cis has higher b.p.
6	All alkynes are acidic	Only TERMINAL alkynes are acidic
7	Lindlar gives trans alkene	Lindlar = CIS; Na/liq.NH ₃ = TRANS
8	Benzene undergoes addition easily	Benzene prefers substitution (preserves aromaticity)
9	C-C in benzene = 133 or 154 pm	139 pm (intermediate — resonance hybrid)
10	Ozonolysis always gives carboxylic acids	Zn/H ₂ O $\rightarrow$ aldehyde/ketone; H ₂ O ₂ $\rightarrow$ acid
11	4n pi = aromatic	(4n+2) pi = aromatic; 4n pi = ANTI-aromatic
12	Hydration of ethyne gives ethanol	Ethyne + H ₂ O $\rightarrow$ Ethanal (not ethanol)
13	cis is non-polar	trans-but-2-ene is non-polar (mu approximately 0)
14	BHC formation is substitution	BHC = addition reaction under UV light
15	Wurtz works for odd C alkanes	Mixed halides give mixture; not selective for odd C
16	Decarboxylation: same C as acid	Product = acid carbon minus 1
17	Br ₂ /CCl ₄ decolourises in substitution	Decolourisation = addition to C=C (unsaturation test)
18	sp ³ C-H most acidic	sp C-H (alkyne) most acidic; sp ³ least acidic
19	1° radical most stable (free radical)	3° radical more stable (like carbocation)
20	EAS: electrophile added to sigma system	EAS: E ⁺ attacks pi cloud $\rightarrow$ arenium ion $\rightarrow$ H ⁺ removed

TOP 20 NCERT HIDDEN FACTS

- 1. Alkanes called "paraffins" = parum affinis (little affinity in Latin)
- 2. Ethene called "ethylene" because it gives oily liquid with Cl₂
- 3. CH₄ found in coal mines and marshy places (marsh gas)
- 4. LPG = liquefied petroleum gas | CNG = compressed natural gas
- 5. Benzene isolated by Michael Faraday in 1825
- 6. Kekule proposed benzene structure in 1865 (after dream of snake biting its own tail)
- 7. X-ray diffraction confirms all 6 C-C bonds in benzene = 139 pm (equal)

- **8. Carbon black (from incomplete combustion) used in ink, printer ink, pigments, filters**
- **9. Petrol/lower petroleum fractions used for dry cleaning (removes grease stains)**
- **10. Like dissolves like: polar in polar; non-polar in non-polar solvents**
- 11. Boiling point: straight chain alkane > branched (more branching = more spherical = less surface area)
- 12. Kharash and Mayo discovered peroxide effect in 1933 at University of Chicago
- 13. Markovnikov proposed his rule in 1869 after studying unsymmetrical alkene reactions
- 14. Torsional strain = repulsion between adjacent C-H bond electron clouds in eclipsed conformation
- 15. Conformations have SAME bond angles and SAME bond lengths (only spatial arrangement changes)
- 16. Staggered and eclipsed represented by Sawhorse or Newman projection
- 17. Polyacetylene films (from ethyne linear polymerisation) used as battery electrodes
- 18. Naphthalene balls: moth repellent; used in toilets (characteristic aromatic smell)
- 19. Benzene burns with sooty/luminous flame (high C:H ratio = 6:6 = 1:1)
- 20. PAHs from incomplete combustion of tobacco, coal, petroleum enter body and damage DNA

TOP 20 EXPECTED NEET QUESTIONS

#	Question Topic	Answer Hook
1	HBr + propene (no peroxide) --> ?	2-Bromopropane (Markovnikov)
2	HBr + propene + peroxide --> ?	1-Bromopropane (anti-Markovnikov)
3	Which gives cis-alkene from alkyne?	Lindlar catalyst
4	Which gives trans-alkene from alkyne?	Na / liquid NH ₃
5	Ozonolysis of 2-methylbut-2-ene --> ?	Propanone + Ethanal
6	Halogens direct to which position?	Ortho/Para (despite deactivating)
7	Which compound is NOT aromatic? (4n or 4n+2)	4n pi compound = not aromatic
8	Most acidic: ethyne, ethene, ethane?	Ethyne (sp, 50% s-character)
9	Kolbe method cannot prepare?	Methane (formate gives CO ₂ only)
10	Most stable conformation of ethane?	Staggered
11	trans vs cis: which has higher m.p.?	trans (better crystal packing)
12	Byproduct in methane chlorination?	Ethane (termination: CH ₃ . + CH ₃ .)
13	Which alkyne does not react with NaNH ₂ ?	Internal alkyne (but-2-yne)
14	Ethyne + H ₂ O (Hg ²⁺ /H ⁺) --> ?	Ethanal (via vinyl alcohol tautomerism)
15	C-C bond length in benzene?	139 pm (intermediate)
16	Which compound undergoes nitration most readily?	Toluene (activating methyl group)
17	BHC formed by?	Addition of Cl ₂ to benzene under UV (free radical)
18	Electrophile in nitration of benzene?	NO ₂ ⁺ (nitronium ion)
19	Assertion-Reason: trans higher m.p. because...?	More symmetrical, better crystal packing
20	Acidity order of ethyne, benzene, hexane?	Ethyne > Benzene > n-Hexane

ALL THE BEST FOR NEET! Master these concepts and score full marks from Hydrocarbons.